Homework 09

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Question 1: Choosing an SVM kernel

I used the nine predictors sbp, tobacco, ldl, adiposity, famhist, typea, obesity, alcohol and age to predict chd. I coded famhist as 1 for Absent and 2 for Present, and made chd a factor.

I compared 18 settings: 3 linear, 9 polynomial and 6 radial. C was 0.01, 1 or 100. Polynomial degree was 2, 3 or 4, with gamma fixed at 1/9. Radial gamma was 0.01 or 1. I held coef0 at 1; degree and gamma are unused for the linear kernel.

I used seed 432 to make five folds, with 93, 93, 92, 92 and 92 observations. Every candidate used these same folds. Within each fold, I calculated the means and standard deviations from training observations only and applied them to the validation observations. I turned off the extra scaling in svm. The reported error is the unweighted mean of the five validation errors.

Kernel	Gamma	C=0.01 (%)	C=1 (%)	C=100 (%)
Linear	-	27.70	28.13	28.13
Polynomial (d=2)	1/9	34.19	27.93	30.10
Polynomial (d=3)	1/9	33.12	31.83	36.57
Polynomial (d=4)	1/9	27.92	33.99	38.95
Radial	0.01	34.63	26.19	27.28
Radial	1	34.63	34.63	37.23

I choose radial, C=1 and gamma=0.01, because it has the lowest mean CV misclassification rate, 26.19%, among these candidates. The next best candidate is radial with C=100 and gamma=0.01 (27.28%). The best linear candidate has C=0.01 and error 27.70%.

The improvement over the best linear candidate is about 1.51 percentage points. This is the result on these five folds. The difference is fairly small, so it does not establish that radial will always perform better. The reproducible R code is included after the skill appendix.

Question 2: SVM analysis report

Data and method. I invoked svm-analysis using saheart.csv, chd as the response, the same nine predictors as Q1, and Absent=1 / Present=2 for famhist. There are 462 observations, nine numeric predictors after coding, and no missing values. The response has 302 zeros and 160 ones. SVM is suitable for comparing linear and nonlinear classification boundaries.

Following the skill, I compared the same 18 candidates using the same five folds as Q1. Scaling used only the training observations in each fold. Polynomial gamma was fixed at 1/9 and coef0 at 1.

Kernel	Gamma	C=0.01 (%)	C=1 (%)	C=100 (%)
Linear	-	27.70	28.13	28.13
Polynomial (d=2)	1/9	34.19	27.93	30.10
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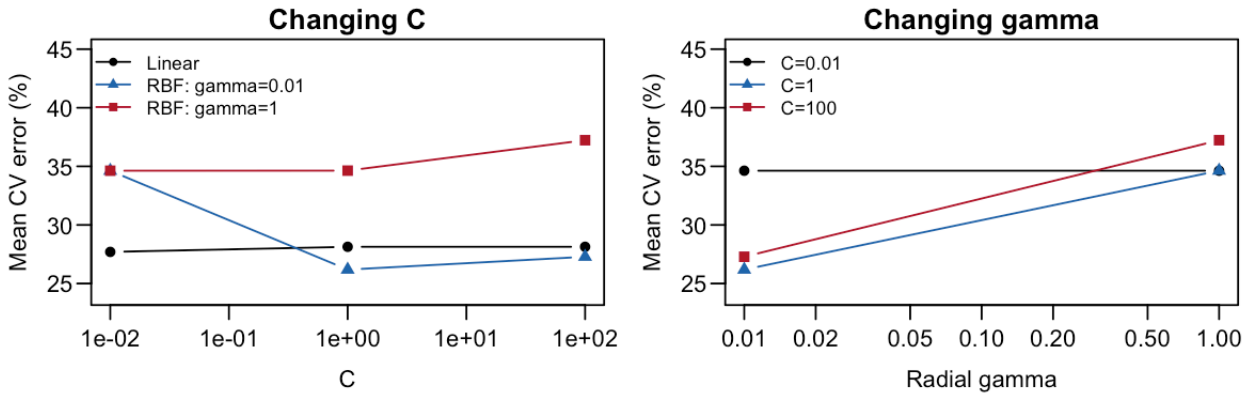


Figure 1: Validation error depends on both C and radial gamma.

Choice and limits. I choose radial with C=1 and gamma=0.01: mean CV error is 26.19%, compared with 27.70% for the best linear model. Larger C did not improve the selected radial model; gamma=1 also gave higher errors. A final refit on all observations uses 307 support vectors. The CV result was used for selection, so it is not an independent test error. The result depends on this fold assignment and the coarse candidate grid, and small differences may not persist on new data.

Appendix: Complete SKILL.md

The following is the complete skill text confirmed before invocation.

```
---
name: svm-analysis
description: Compare SVM models and choose the best settings using cross-validation.
---

Use this skill only when I explicitly ask for it.

Check if there are missing values and whether the response is categorical.
Decide if the data is suitable for SVM.

Compare linear, polynomial and radial kernels.
Linear is a simple baseline, while polynomial and radial can fit nonlinear relationships.

Tune C for all kernels, degree for polynomial, and gamma for radial.
Use C and gamma values spaced by factors of 10 or 100.
State other settings held fixed.

Use the same five folds for all candidates.
Standardize using only training data in each fold and apply the same scaling to validation data.
Choose the model with the lowest mean CV misclassification rate.

Include a compact table showing kernels, parameter settings and mean CV errors.
Include a plot showing how CV error changes with C and gamma.

Report the final kernel, parameters and mean CV error.
Explain the choice and limitations.
Keep the report within one printed page at 11-point font.
```

Appendix: Question 1 R code

The following code generates all 18 candidate CV errors. Polynomial gamma=1/9 and coef0=1 are held fixed. Numeric degree/gamma placeholders for unused kernel arguments do not change those kernels.

```
df <- read.csv("data/saheart.csv")
df$famhist <- ifelse(df$famhist == "Absent", 1, 2)
df$chd <- factor(df$chd)
x <- df[, 1:9]; y <- df$chd
set.seed(432); fold <- sample(rep(1:5, length.out=nrow(df)))
library(e1071); costs <- c(0.01, 1, 100); gammas <- c(0.01, 1)
cv_error <- function(kernel, cost, degree, gamma) {
    errors <- numeric(5)
    for(i in 1:5) {
        train <- fold != i; valid <- fold == i
        m <- colMeans(x[train, ]); s <- apply(x[train, ], 2, sd)
        x_train <- scale(x[train, ], center=m, scale=s)
        x_valid <- scale(x[valid, ], center=m, scale=s)
        fit <- svm(x=x_train, y=y[train], kernel=kernel, cost=cost, degree=degree, gamma=gamma,
coef0=1, scale=FALSE)
        pred <- predict(fit, newdata=x_valid)
        errors[i] <- mean(pred != y[valid])
    }
    return(mean(errors))
}

linear <- data.frame(kernel="linear", cost=costs,
                     degree=1, gamma=1/9)
poly <- expand.grid(kernel="polynomial", cost=costs,
                    degree=2:4, gamma=1/9,
                    stringsAsFactors=FALSE)
radial <- expand.grid(kernel="radial", cost=costs,
                      degree=1, gamma=gammas,
                      stringsAsFactors=FALSE)
candidates <- rbind(linear, poly, radial)
candidates$error <- numeric(nrow(candidates))
for(j in 1:nrow(candidates)) {
    candidates$error[j] <- cv_error(
        kernel=candidates$kernel[j],
        cost=candidates$cost[j],
        degree=candidates$degree[j],
        gamma=candidates$gamma[j]
    )
}
best <- which.min(candidates$error)
candidates[best, ]
```